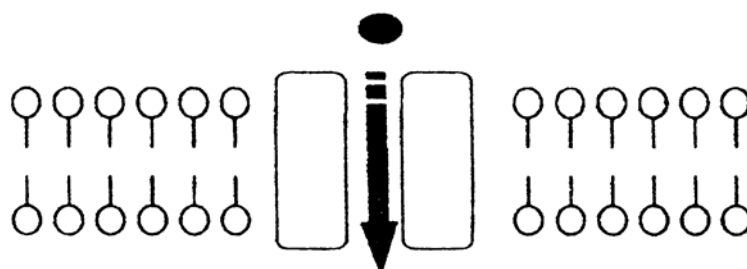


- 1 The table gives descriptions of four membranous structures in a cell. Which structure is correctly matched with its function?

	Structure	Function
A	An extensive network of tubes and sacs; each tube and sac bounded by a single membrane	lipid synthesis
B	A spherical sac bounded by a single membrane	protein synthesis
C	A sac bounded by two membranes, the inner highly folded	packaging of proteins
D	A stack of elongated, curved sacs; each sac bounded by a single membrane	photosynthesis

- 2 The diagram below shows the pathway of a particle through a cell surface membrane.



What are the likely properties of the particle?

	Relative molecular mass	Electrically charged
A	High	Yes
B	Low	No
C	High	No
D	Low	Yes

- 3 High concentrations of urea break all bonds, except covalent bonds, in protein molecules. Which level of protein structure would remain unchanged when keratin is treated with urea?
- A** Primary
- B** Secondary
- C** Tertiary
- D** Quaternary

4 Which formula is a carbohydrate?

A CH_3CHO

B $\text{C}_2\text{H}_5\text{OH}$

C $\text{C}_3\text{H}_6\text{O}_3$

D $\text{C}_{15}\text{H}_{31}\text{COOH}$

5 Which of the followings is are / true about cellulose and collagen?

1 Both serve as structural components as they have high tensile strength.

2 The high tensile strength is a result of interchain hydrogen bonds

3 Both show quaternary level of organization.

4 Cellulose molecules can only form hydrogen bonds whereas collagen is capable of forming other bond in addition to hydrogen bonds.

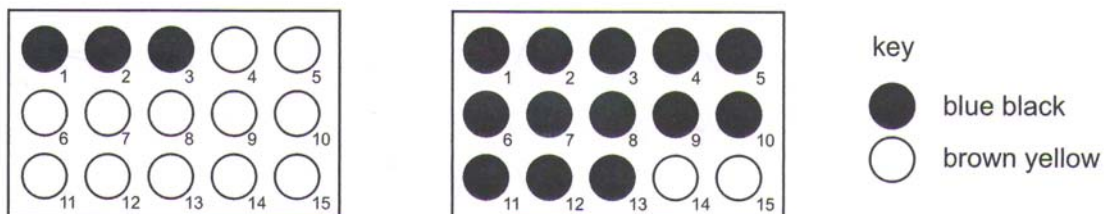
A (1) only

B (1) and (2) only

C (1), (2) and (3) only

D All of the above

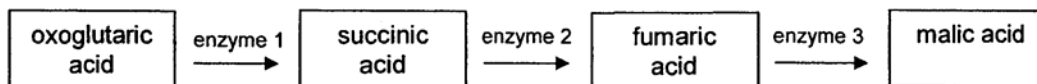
- 6 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which shows the correct temperatures and times for the complete digestion of starch?

	Temperature/ °C	Time/s
A	30 10	3.15 0.45
B	30 10	45 195
C	10 30	45 195
D	10 30	3.15 0.45

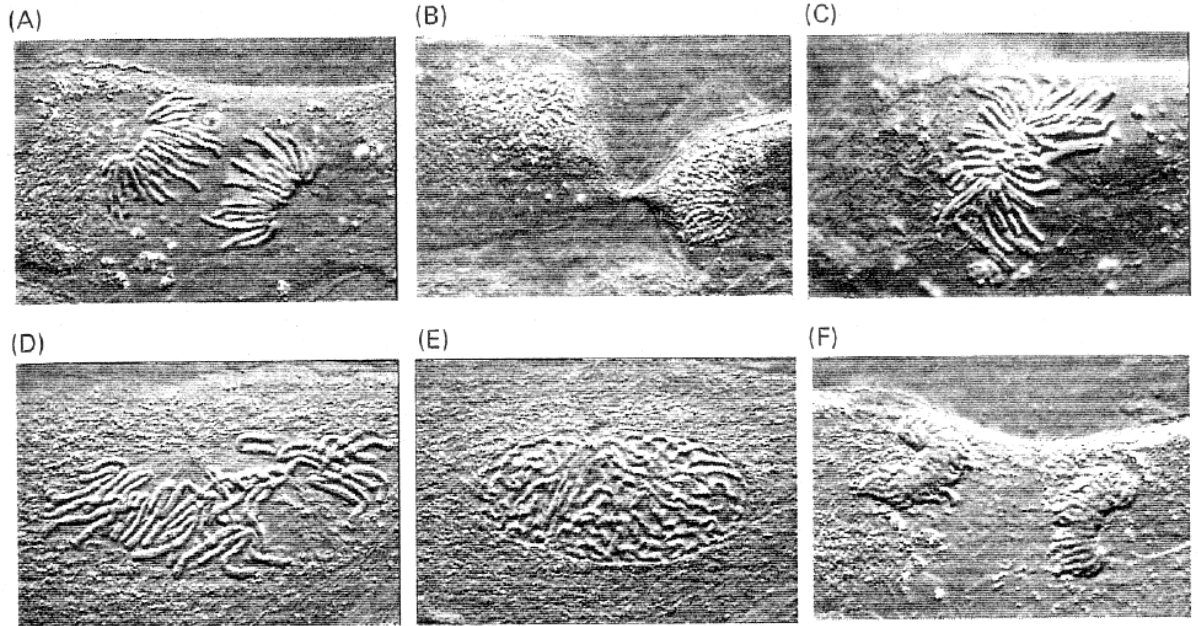
- 7 The four acids shown below form part of an enzyme-catalyzed sequence.



The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid. Further addition of fumaric acid results in the formation of malic acid. What does this information indicate about malonic acid?

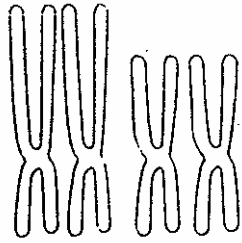
- A** It is an inhibitor of enzyme 2.
- B** It catalyzes the formation of succinic acid.
- C** It is an inhibitor of enzyme 1.
- D** It reacts with fumaric acid.

- 8 A living cell from the lung epithelium of a newt is shown at different stages of mitosis in the micrographs (**P** to **U**) below. Identify the correct sequence of these micrographs.

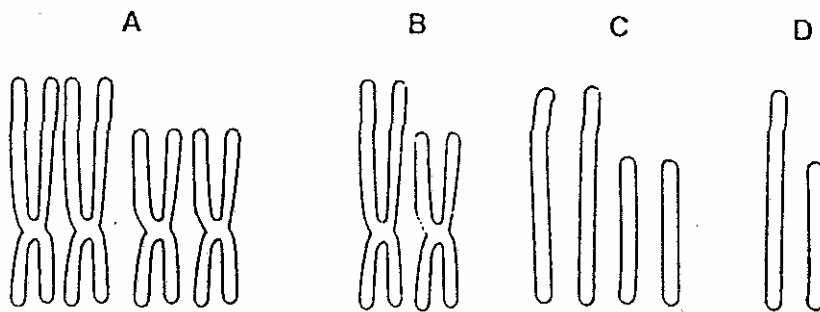


- A ECDAFB
- B DECAFB
- C EDCAFB
- D EDCBAF

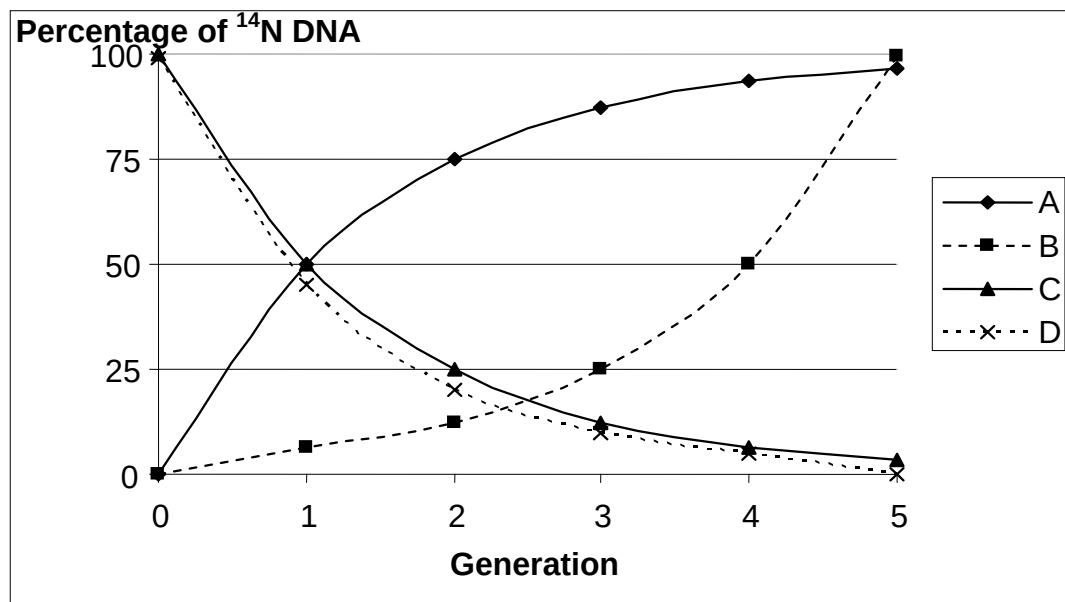
- 9 The diagram shows the chromosome complement of a cell just before the metaphase stage of mitosis.



What will be the chromosome complement in telophase?



- 10 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for 5 generations. The percentage of ^{14}N DNA strands in each generation was estimated. Which curve provides evidence that DNA replication is semi-conservative?



- 11** Five different amino acids (numbered 1 – 5 below) form the following sequence in part of a polypeptide chain:

1 - 2 - 3 - 4 - 2 - 5 - 3

Messenger RNA (mRNA) codons which correspond to these amino acids are:

Amino acid 1: UGU

Amino acid 2: GAU

Amino acid 3: CAC

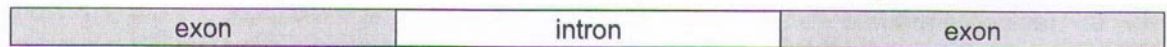
Amino acid 4: UAG

Amino acid 5: AAG

Which one of the following DNA base sequences could provide the code for the given section of polypeptide?

- A** ACACTTGTGATGCTATTCGTG
 - B** ACACUAGUGAUGCUAUUCGUG
 - C** ACACTAGTGATGCTAAACGTG
 - D** ACACTAGTGATCCTATTCGTG
- 12** Which statement could NOT describe both eukaryotic and prokaryotic transcription?
- A** Promoter elements in the DNA are required for polymerase binding.
 - B** Transcription factors control the binding of RNA polymerase.
 - C** RNA polymerase opens a double stranded DNA to expose the template strand.
 - D** RNA polymerase catalyzes RNA synthesis from 3' to 5' of the new strand.
- 13** Telomerase activity in humans is generally restricted to the germ line, cancerous cells, and a few specific adult cell types. How would this be expected to impact humans?
- A** Cancer cells can divide indefinitely.
 - B** Because there is a difference in number of divisions, sperm would be expected to have shorter telomeres than egg cells.
 - C** Rapidly dividing populations of cells have telomeres equal in size to slowly dividing populations of cells.
 - D** None of these.

- 14 The diagram shows a gene in a human cell.

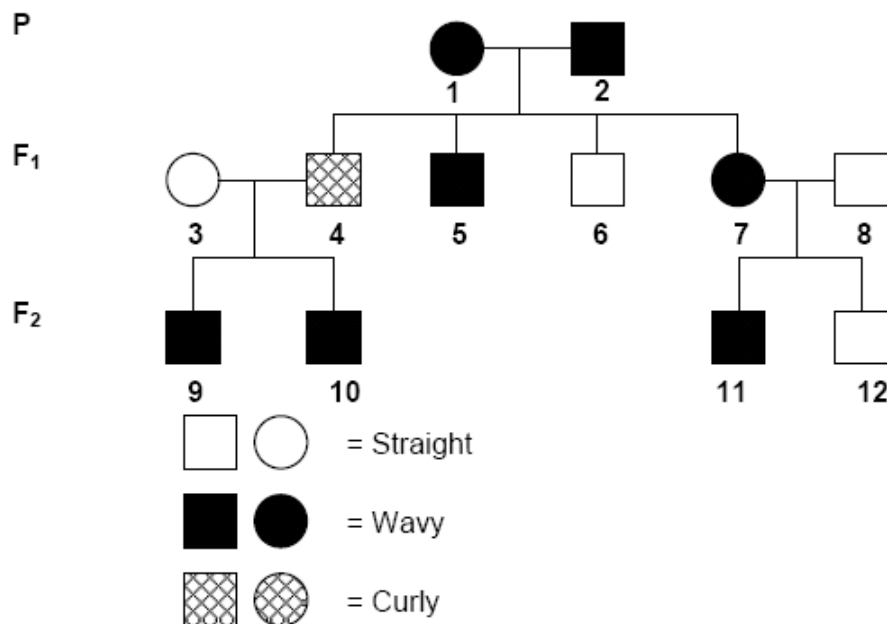


In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

- A Activation of the gene by a steroid hormone
- B Attachment of mRNA to a ribosome
- C Binding of RNA polymerase to DNA
- D Splicing of pre-mRNA

Questions 15 and 16 refer to the following pedigree chart.



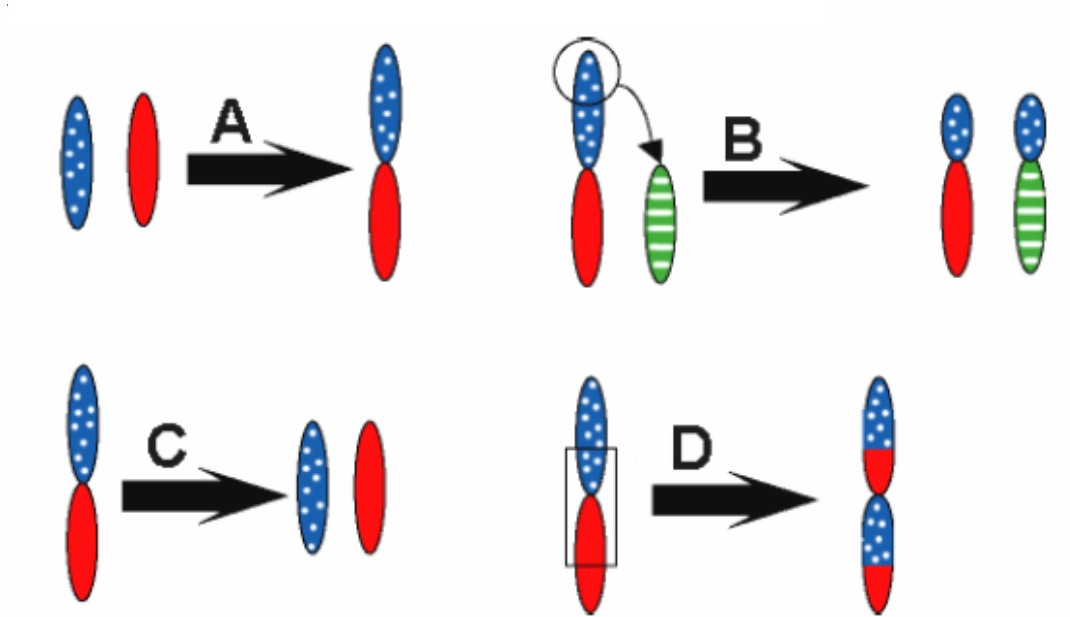
- 15 Which of the following statements is true?

- A Wavy hair is dominant.
- B Individual 1 is homozygous.
- C Individual 4 is homozygous.
- D Curly hair is recessive to straight hair.

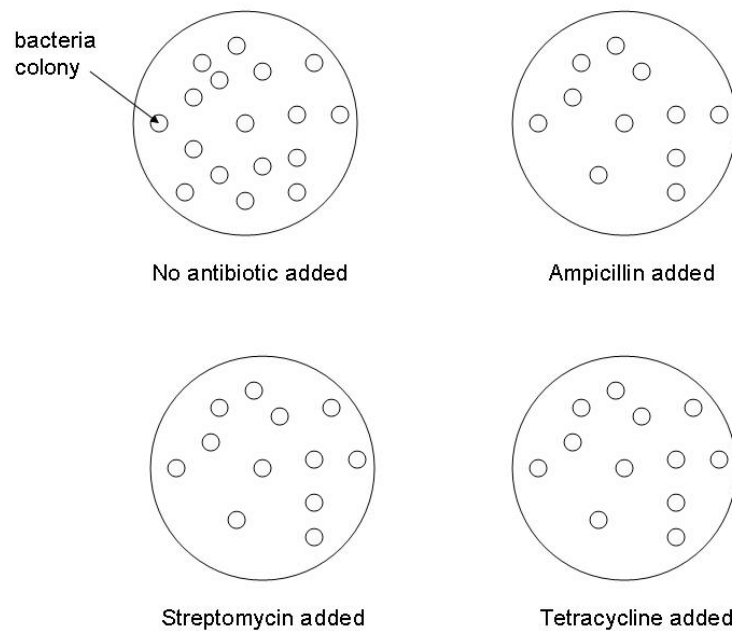
16 Which of the following statements has a 50% chance of being true?

- I** One of the parents of individual **2** had the same phenotype as individual **2**.
 - II** If individual **10** married someone with wavy hair, the first child would have wavy hair.
 - III** If individual **6** married a woman with straight hair, all of the offspring would have straight hair.
- A** II only
- B** I and III only
- C** II and III only
- D** All of the above

17 Which of the following illustrates chromosomal inversion?



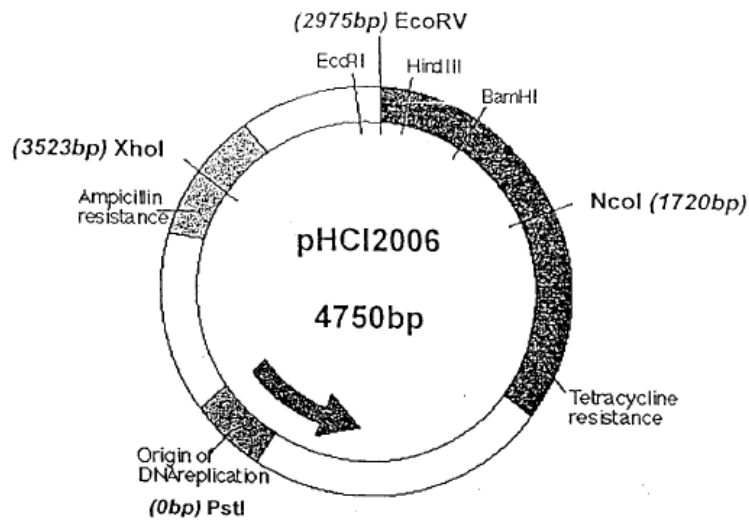
- 18** The DNA containing the gene coding for insulin and plasmid with genes coding for resistance against tetracycline, ampicillin and streptomycin were mixed together with the appropriate restriction enzymes. A heat shock treatment was applied to bacteria to allow the plasmids to enter the cells. The bacteria was plated and grown in the following media.



Looking at the diagram above, where is the gene coding for insulin inserted?

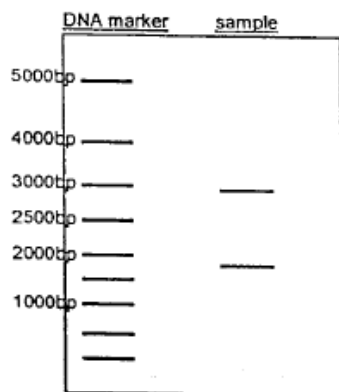
- A** Ampicillin
- B** Streptomycin
- C** Tetracycline
- D** Ampicillin and Tetracycline

- 19 A gene Y, 726bp in size, is inserted into the vector, pHC12006, using *XhoI*.

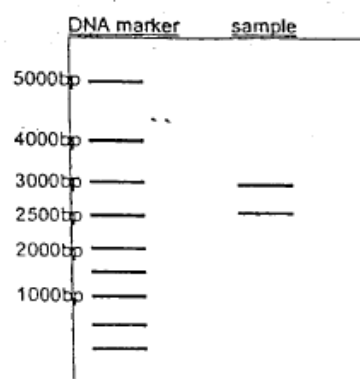


The recombinant DNA is then digested with *EcoRV* and *PstI* and separated by size through gel electrophoresis. The sizes of the separated strands are determined by comparison to a DNA marker. Which of the following gel electrophoresis diagrams shows the correct sizes of the separated strands?

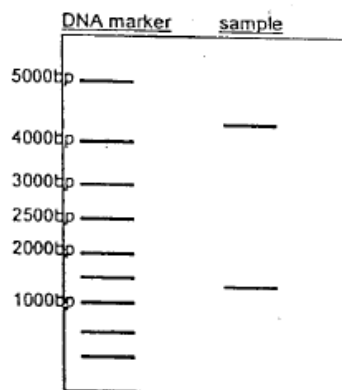
A



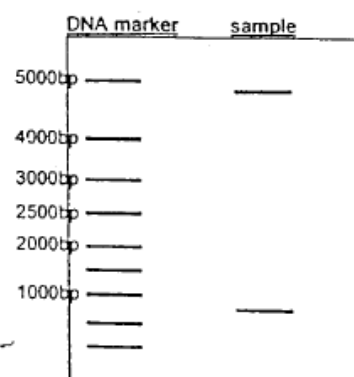
B



C



D



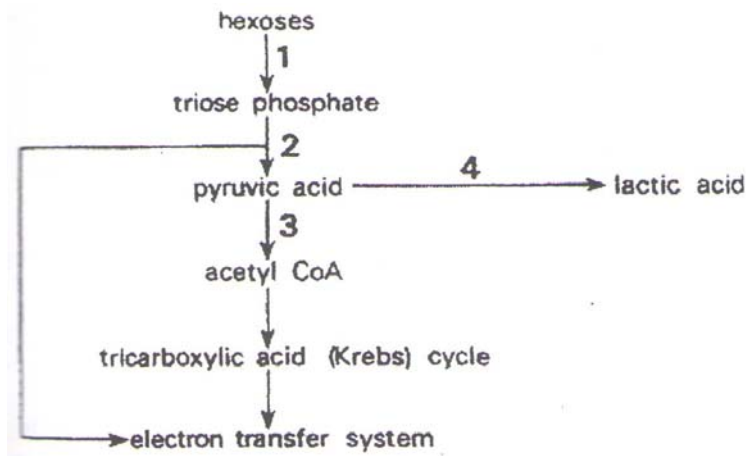
- 20** Which of the following would *not* be true of the polymerase chain reaction?
- A** It could be used to amplify a specific target segment of DNA
 - B** It consists of a three-step cycle involving DNA denaturation, annealing and extension.
 - C** It has an advantage over cell-based cloning due to its sensitivity and accuracy.
 - D** It requires a heat-stable DNA polymerase to extend the primers.
- 21** Which statement describes a unique feature of stem cells in adult humans?
- A** They can differentiate into different cell types.
 - B** They can undergo meiosis.
 - C** They have a complete human genome.
 - D** They have multiple copies of active genes.
- 22** Liposuction waste has been reported to contain 50-100 million stem cells per 250 g. When these stem cells were harvested, they could not differentiate into any other cell except for fat, bone and cartilage.
- Which of the following statements best explains this?
- A** It is a multipotent adult stem cell which can only specialise into limited cell types.
 - B** It is a multipotent embryonic stem cell which can specialise into almost all the cell types.
 - C** It is a pluripotent embryonic stem cell which can specialise into almost all the cell types.
 - D** It is a pluripotent adult stem cell which can specialise into almost all the cell types.
- 23** It has been found that an aqueous suspension of isolated chloroplasts will evolve oxygen if illuminated in the presence of a certain type of compound.

Which type of compound and which colours of light are required for maximum oxygen evolution?

	Type of compound	<i>Colours of light at which maximum evolution occurs</i>
A	Electron acceptor	Blue and green
B	Electron acceptor	Blue and red
C	Electron donor	Blue and green
D	Electron donor	Blue and red

- 24** Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternative route for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle (dark reaction) be affected in chloroplasts poisoned with dinitrophenol?
- A** No effect since Calvin cycle is an enzyme-controlled process.
 - B** The rate of Calvin cycle will increase because pH in the stroma will drop towards the optimum.
 - C** The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate (GP).
 - D** The rate of Calvin cycle will decrease with the accumulation of glyceraldehyde-3-phosphate (G3P) i.e. triose phosphate (TP).
- 25** Which two reactions are involved in the process of photophosphorylation?
- A** hydrolysis of ATP and the reduction of GP (PGA)
 - B** hydrolysis of ATP and the reduction of NADP
 - C** synthesis of ATP and the oxidation of reduced NADP
 - D** synthesis of ATP and the reduction of NADP

- 26 The diagram summarises the stages involved in respiratory metabolism of mammalian skeletal muscle.

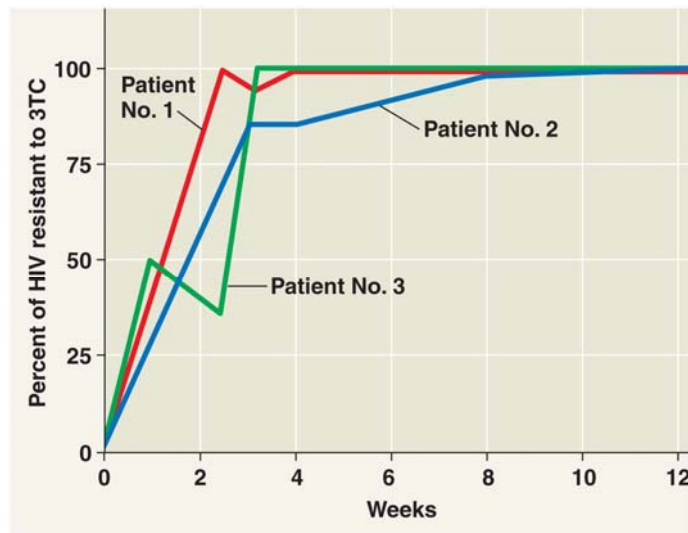


Stage 4 in the diagram occurs in mammalian muscle cells under anaerobic conditions. Why is stage 4 necessary?

- A to detoxify pyruvate
 - B to form lactate which is immediately reconverted to glycogen
 - C to oxidise the reduced NAD^+ facilitating stage 2
 - D to reduce the oxidised NAD^+ facilitating stage 3
- 27 When cyanide is bound to cytochrome oxidase, the cell can no longer aerobically produce ATP for energy. Which statement below best explains this?
- A It prevents photolysis from occurring to produce oxygen.
 - B It prevents NADPH from being oxidised to NADP^+ , hence preventing electron flow down the electron transport chain. No chemiosmosis occurs.
 - C It prevents pyruvate from being synthesised in glycolysis, hence stopping the Krebs cycle.
 - D It prevents oxygen from accepting electrons and protons to form water, hence preventing electron flow down the electron transport chain. No chemiosmosis occurs.

- 28 A young boy never had much energy. He goes to a doctor for help and is sent to the hospital for some tests. There they discover his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal. Of the following, which is the best explanation of his condition?
- A His cells cannot move NADH from glycolysis into the mitochondria.
 - B His mitochondria lack the transport protein that moves pyruvate across the outer mitochondrial membrane.
 - C His cells contain something that inhibits oxygen use in his mitochondria.
 - D His cells have a defective electron transport chain, so glucose goes to lactate instead of to acetyl CoA.
- 29 Patients were treated with 3TC, a drug that combats HIV. 3TC resembles the cytosine-bearing nucleotide. HIV's reverse transcriptase picks up a 3TC molecule instead of a C-bearing nucleotide and inserts the 3TC into a growing DNA chain. This error terminates further elongation of the DNA and thus blocks reproduction of HIV.

3 patients were given 3TC but after an initial drop in the concentration of HIV in the patient, the concentration of HIV increased. It was found that the drug is no longer effective in the treatment of HIV.



Which statement gives the best explanation for this?

- A 3TC causes HIV to mutate into resistant forms.
- B Some HIV possess inheritable resistance to 3TC.
- C The patients have become resistant to the effects of 3TC.
- D Continued use of 3TC has gradually increased the resistance of human populations to HIV.

- 30** Molecular homology of haemoglobin is used to determine evolutionary relatedness in a range of mammals. Either nucleic acids or proteins can be sequenced. The sequence of nucleic acids and proteins are compared.

Which sequence will vary **least** in related mammals?

- A** Amino acid
- B** DNA
- C** Edited mRNA transcript
- D** Primary mRNA transcript